TRIBHUVAN UNIVERSITY

Institute of Science and Technology

A

Final Project Report On

E-Shopp – An online E-Commerce Application

In the regular fulfillment of the requirements for the Bachelors of Science in Computer Science and Information Technology (BSc CSIT)

Submitted to:

Department of Science and Technology

National Infotech College

Submitted by

Ruby Shah [29849/078]

August, 2025

Under the Supervision of

Er. Amrendra Chaurasia

Tribhuvan University

Institute of Science and Technology

National Infotech College

Supervisor’s Recommendation

I hereby recommend that this project prepared under my supervision by Ruby Shah entitled “E-Shopp” in the partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Information Technology is recommend for the final evaluation.

____

SIGNATURE

Er. Amrendra Chaurasia

SUPERVISOR

National Infotech College

Institute of Science and Technology

Tribhuwan University

Institute of Science and Technology

National Infotech College

Letter of Approval

This is to certify that this project prepared **Ruby Shah** entitled "E-Shopp – An Online E-Commerce Application" in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Information Technology. has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

SIGNATURE of Supervisor

SIGNATURE of Coordinate

Signature of Internal Examiner

Internal Examiner

Signature of External Examiner

External Examiner

ACKNOWLEDGEMENT

Presentation, inspiration and motivation have always played a key role in the success of any venture. We sincerely acknowledge and thank those who have contributed their valuable time in helping us to achieve success in our project work. We would like to express our gratitude to all those who gave us the possibility to complete this project. We want to thank **National Infotech College** for giving us the opportunity for doing this project.

We are indebted and thankful to our Project Guide **Er. Amrendra** Chaurasia to whom we owe his piece of knowledge for his valuable and timely guidance. We would also like to thank our coordinator **Mr. Bijay Kushwaha** and project supervisor **Er. Amrendra Chaurasia** for their assistance and insightful comments, and who willingly shared their expertise with us.

Last, but not the least, our parents are also an important inspiration for us. So, with due regards, we express our gratitude to them.

Ruby Shah [29849/078]

(BSc. CSIT, 4th Years / 7th Semesters)

ABSTRACT

In today's digital age, the internet plays a vital role in connecting people to products, services, and opportunities. As people's demand for a better quality of life and increased convenience grows, online platforms have become essential in fulfilling everyday needs. However, many local businesses and sellers still struggle to reach a broader market due to the lack of proper digital infrastructure or visibility.

"E-Shopp" is an e-commerce web application designed to address this issue by bridging the gap between local sellers and customers, with an initial focus on Birgunj, Nepal. This platform allows businesses and individual vendors to list their products under specific categories, making them accessible to a wider customer base. An admin reviews product listings before they are published, ensuring quality and accuracy.

E-Shopp serves as a localized online marketplace, aiming to boost local commerce by streamlining the buying and selling process. Developed using HTML, Tailwind CSS, JavaScript, Bootstrap, Firebase and fire Store, the platform provides a user-friendly experience that simplifies product browsing, ordering, and transaction management for both customers and sellers.

Keywords: *E-commerce, Online Shopping, Nepali Market, E-Shopp, Local Business Platform*

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LIST OF TABLES

Table No:	Table Name:	Description:
Table 1	Admin	Stores information about platform administrators
Table 2	Categories	Contains product categories (e.g., electronics, clothing)
Table 3	Sellers / Shops	Stores information about individual sellers or companies
Table 4	Products	Contains details of products listed for sale
Table 5	Orders	Stores order records placed by customers
Table 6	Users / Customers	Contains data of registered users (customers)
Table 7	Login Testing	Stores results or logs from login functionality testing

LIST OF FIGURES

Figure No.	Figure Title	Description
Figure 1	Use Case Diagram	Diagram showing interactions between users and the system
Figure 2	Gantt Chart	Project timeline with tasks and milestones
Figure 2	ER Diagram	Entity-Relationship diagram showing database structure
Figure 3	Level-0 DFD	High-level Data Flow Diagram of E-Shopp
Figure 4	Level-1 DFD	Detailed breakdown of main processes in the system
Figure 5	Level-2 DFD (Order Management)	Data flow for order placement, processing, and tracking
Figure 6	Database Schema Diagram	Visual representation of tables and their relationships
Figure 6	Login Page	Design of the login interface for customers

LIST OF ABBREVIATION

Abbreviation	Full Form
BSc. CSIT	Bachelor of Science in Computer Science and Information Technology
Firebase	Firebase
HTML	Hypertext Markup Language
Tailwind CSS	Tailwind Cascading Style Sheets
JS	JavaScript
API	Application Programming Interface
DB	Database
UAT	User Acceptance Testing
ER	Entity-Relationship (used in database design)
DFD	Data Flow Diagram (used in system analysis)
CRUD	Create, Read, Update, Delete (common database operations)

Chapter 1: Introduction

1.1 Introduction

"E-Shopp" is an online e-commerce web application designed to bridge the gap between local businesses and digital consumers, especially in regions like **Birgunj, Nepal**, where small and medium sellers struggle with limited online presence. Built **with HTML, Tailwind CSS, JavaScript, Bootstrap**, and powered by **Firebase** on the backend, E-Shopp allows vendors to register and sell their products directly to customers online.

Customers can browse a categorized list of products, view details, add items to a shopping cart, and place orders with ease. Vendors (sellers) can manage their product listings, view order requests, and handle fulfilment through a user-friendly dashboard. Each user—whether customer or seller—has a personalized experience based on their role.

E-Shopp is designed to support the local economy by enabling digital commerce for smaller sellers and offering customers a simplified and trustworthy way to shop for local goods online. With real-time database integration, authentication, and responsive design, the application provides a modern and accessible platform for digital transactions.

1.2 Problem Statement

In many local areas such as **Birgunj**, there is a lack of accessible digital platforms for small vendors to sell their products online. Traditional marketplaces are either unavailable or too complicated and expensive for smaller sellers. Likewise, customers struggle to discover and purchase local products efficiently. E-Shopp aims to resolve this by creating a streamlined e-commerce platform that is easy to use, scalable, and tailored for local needs.

1.3 Objectives

The key objectives of E-Shopp are:

- To develop an accessible online shopping platform for local sellers and buyers.
- To provide a clean, modern user interface using frontend technologies like HTML, Tailwind CSS, JS, and Bootstrap.
- To utilize Firebase for real-time database, authentication, and product storage.
- To empower local sellers by giving them an online presence and access to wider markets.
- To simplify the shopping process for users, from browsing to checkout.

1.4 Scope and Limitations

a. Scope:

- Enables local sellers to create digital storefronts and manage their inventory.

- Supports core e-commerce functionalities: product listing, cart management, order processing, and user authentication.
- Designed with responsiveness in mind to work across mobile and desktop.
- Focused initially on Birgunj, with plans to scale across Nepal.
- Allows real-time interactions using Firebase.

b. Limitations:

- Internet access is required to use the platform.
- Limited payment methods in the initial phase (e.g., no payment gateway integration).
- Currently lacks multi-language support.
- Product variety and availability depend on seller participation.
- Firebase's free tier has limits which may affect scaling if not upgraded.

1.5 Development Methodology

Prototyping methodology was adopted for the development of E-Shopp. This iterative model enabled rapid development, user feedback, and continuous improvements.

Steps:

1. **Requirement Gathering**
    - Defined core features: user login, product listing, shopping cart, order handling.
2. **Wireframe and Low-Fidelity Prototype**
    - Designed initial layouts and structure using Figma.
3. **Feedback and Design Improvements**
    - Collected feedback from peers and mentors to improve UI/UX.
4. **Iterative Development**
    - Frontend components built with Tailwind CSS and Bootstrap.
    - Firebase configured for database and authentication.
5. **High-Fidelity Prototype**
    - Developed realistic interactive flows for seller dashboard, product view, and customer cart.
6. **Final Implementation**
    - Integrated all modules into a complete, functional e-commerce system.

1.6 Report Organization

The structure of this report for E-Shopp is as follows:

a. Title Page

- **Title of the Project: E-Shopp** – A Web-Based Local E-commerce Platform

- **Author**: Ruby Shah
- **Degree:** BSc. CSIT
- **Academic Year**: 2082
- **Institution:** National Infotech College

b. Abstract

- Summary of aims, tools used (HTML, Tailwind CSS, Firebase), outcomes, and project relevance.

c. Table of Contents

- Organized section headers and page numbers.

d. Acknowledgments

- Credits to mentors, contributors, peers, and advisors.

e. List of Figures and Tables

- Lists all visual and data elements used in the report.

f. Introduction

- Overview of E-Shopp, the challenges it addresses, and its purpose.

g. Literature Review

- Existing e-commerce platforms (e.g., Daraz, Amazon) and their limitations in local context.
- Identifies how E-Shopp fills the gap for local vendors.

h. Project Scope and Objectives

- Defines geographic and technical boundaries.
- Outlines expected outcomes and features.

i. Methodology

- Description of prototyping methodology.
- Step-by-step development and user-centered design approach.

j. Requirement Analysis

- Functional requirements (e.g., add to cart, product upload).
- Non-functional requirements (e.g., scalability, security).
- User stories and use cases.

k. System Design

- Firebase architecture, data structure (NoSQL)..
- Component interaction diagrams and user flow.

l. Implementation and Development

- Technologies used:

Frontend: HTML, Tailwind CSS, JS, Bootstrap

Backend: Firebase Authentication, Fire store Database

- Code snippets and feature development process.

m. Testing

- Unit tests for components.
- Integration tests across workflows (e.g., login → cart → checkout).
- Bug tracking and resolution.

n. Conclusion

- Summary of outcomes, effectiveness of the app, and real-world application potential.

o. Recommendations

- Future features:
- ▪ Online payments integration (e Sewa, Khalti)
- ▪ Multilanguage support
- ▪ AI-driven product recommendations
- ▪ Progressive Web App (PWA) version

p. References

- All citations and tools used, formatted using **IEEE Style**.

q. Appendices

- Source code snippets
- Firebase database structure
- Additional diagrams (e.g., ER)

Chapter 2: Background Study and Literature Review

2.1 Background Study

The rise of the internet and digital technologies has revolutionized the way consumers shop and businesses sell their products. Global platforms like **Amazon, eBay**, and **Alibaba** have made e-commerce an essential part of modern life. However, these platforms often focus on international or national markets and do not cater specifically to local vendors in small towns or developing regions.

In areas like **Birgunj, Nepal,** many small businesses and individual sellers still rely on traditional methods such as local shops, word-of-mouth promotion, and occasionally social media (e.g., Facebook or WhatsApp) to promote their products. These approaches are limited in reach and inconsistent in effectiveness, making it difficult for small sellers to grow or compete with larger businesses.

At the same time, local consumers lack a centralized, reliable platform to browse and purchase locally available products. They may be unaware of nearby sellers offering what they need or may find it inconvenient to visit stores in person.

As **internet access** and **smartphone usage** continue to rise in Nepal—especially among the younger population—there is an increasing need for an **easy-to-use, localized e-commerce platform. E-Shopp** is developed to meet this demand by offering **a user-friendly, responsive web application** that allows sellers to list their products and customers to browse and order them conveniently.

Using modern tools like **HTML, Tailwind CSS, Bootstrap, and JavaScript** for the frontend, and **Firebase** for real-time database and authentication, E-Shopp ensures a seamless experience for both sellers and buyers. The platform empowers local businesses and enhances community-level commerce by making product discovery, purchase, and sales management digital and accessible.

2.2 Literature Review

1. E-Commerce and Local Market Digitization

Research shows that e-commerce platforms significantly improve market access for both consumers and sellers. According to **Patel et al. (2019),** digital commerce enhances product visibility, expands customer base, and improves business scalability. However, large-scale platforms like Amazon or Flipkart are often too complex or expensive for small, local vendors to use effectively. **E-Shopp** seeks to fill this gap by offering a lightweight, no-cost solution tailored to local sellers in Birgunj.

2. Localized Platforms in Developing Economies

A study by **Shrestha & Bista (2021**) on digital platforms in Nepal concluded that **localized platforms see better adoption and long-term engagement** than generalized or global solutions. For example, apps targeting local agriculture, education, or health services were found to be more effective when they reflected local needs, language, and culture. E-Shopp adopts a similar localization strategy, helping Birgunj-based sellers reach nearby consumers in a familiar, culturally relevant digital environment.

3. Firebase in Modern Web Applications

According to **Google Developer Reports (2022), Firebase** is increasingly popular for building scalable, real-time web applications due to its ease of integration, real-time database, secure authentication, and serverless architecture. Firebase's NoSQL Fire store database and built-in Authentication system allow developers to rapidly build and deploy functional e-commerce solutions. E-Shopp utilizes Firebase to manage users, product data, and orders without needing a traditional server, reducing complexity and cost for local deployment.

Summary

Existing e-commerce literature emphasizes the importance of:

- Localized solutions for better user adoption,
- Real-time backend technologies like Firebase for simplicity and speed,

E-Shopp combines all of these elements to serve as a **community-first digital marketplace** that empowers both local consumers and small business owners in Birgunj, while remaining scalable for future growth across Nepal.

Chapter 3: System Analysis

3.1. System Analysis

3.1.1 Requirement Analysis

i. Functional Requirements

This section outlines the main features of the system categorized by user roles.

a. Admin (optional role for platform moderation)

- Manage seller and customer profiles
- Monitor reported products or users
- Review listed products for quality control
- Control access and assign user roles (if needed)
- View platform usage reports

b. Seller

- Register/Login with Firebase Authentication
- Add new product (title, description, price, image, category)
- Edit or delete their own product listings
- View/manage orders from customers
- Manage inventory (stock level updates)
- Upload store logo and customize store profile

c. Customer/User

- Register/Login
- Browse products by category, keyword, or search filter
- Add products to cart
- Place an order
- View order history
- Manage personal profile (name, address, contact info)

d. Notification System

- Customers receive real-time notifications on order status updates
- Sellers receive notifications when an order is placed
- Admin (if implemented) gets alerts for reported products

Use Case Diagram

(Diagram should include actors: Admin, Seller, Customer, and key interactions like "Add Product," "Place Order," "Receive Notification.")(Can be generated upon request.)

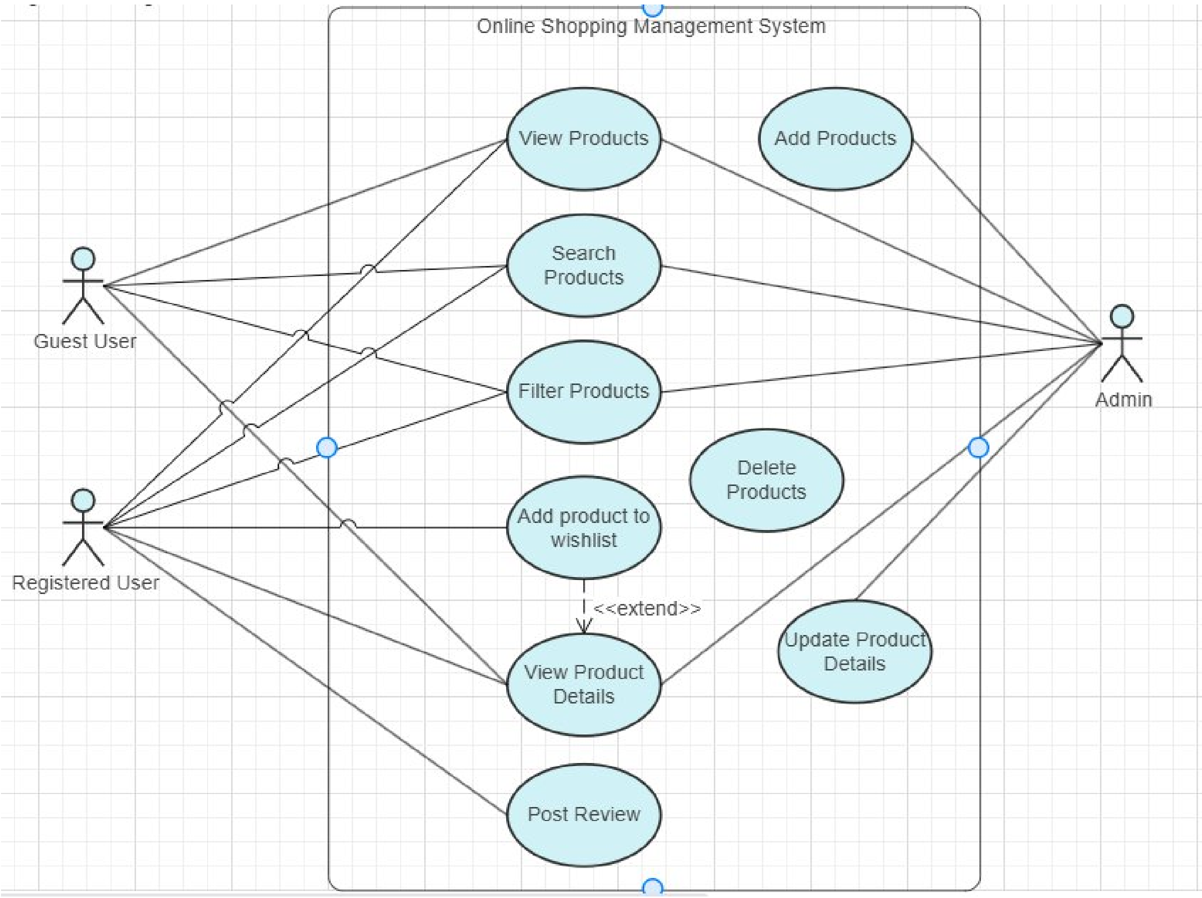

Figure 1: Use Case Diagram for E-Shopp

ii. Non-Functional Requirements

- **Usability:** Clean and mobile-friendly UI using Tailwind CSS & Bootstrap
- **Performance**: Fast load time using Firebase for real-time data sync
- **Security:** Firebase Authentication ensures secure access; Fire store rules protect data
- **Availability:** Platform is hosted online and accessible 24/7
- **Responsiveness:** Adapts smoothly across devices (mobile, tablet, desktop)
- **Scalability:** Firebase backend supports real-time scaling

3.1.2 Feasibility Analysis

i. Technical Feasibility

E-Shopp is built using well-supported technologies:

- **Frontend:** HTML, Tailwind CSS, JavaScript, Bootstrap
- **Backend:** Firebase (Fire store, Authentication, Hosting)
- These technologies reduce backend complexity, support scalability, and accelerate development.

ii. Operational Feasibility

The application is intuitive for both sellers and buyers. A clean layout, minimalistic design, and responsive behaviour ensure ease of use even for non-tech-savvy users in areas like Birgunj. Sellers can easily manage their store, and buyers can browse or order in just a few clicks.

iii. Economic Feasibility

The project uses Firebase's free tier and open-source technologies, which keeps development and operational costs low. Hosting is included in Firebase, eliminating the need for external hosting services initially.

iv. Schedule Feasibility

A 12-week development plan is proposed using an agile methodology. Tasks are broken down as follows:

Key Activities	1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	11th
Planning	✔	✔									
Analysis & Design		✔	✔	✔							
Coding			✔	✔	✔	✔	✔	✔			
Testing						✔	✔	✔	✔	✔	
Evaluation									✔	✔	✔
Documentation								✔	✔	✔	✔
Deployment										✔	✔
Final Presentation											✔

Figure 2: Gantt Chart – E-Shopp Development Timeline

3.1.3 Analysis

If Structured approach:

a. Data Modelling using ER Diagram for E-Shopp

The ER Diagram for E-Shopp consists of the following entities:

- **User**
  - o User ID, Name, Email, Role (Customer/Seller), Address
- **Product**
  - o Product ID, Title, Price, Image URL, Stock, Seller ID
- **Cart**
  - o Cart ID, User ID, Product ID, Quantity
- **Order**
  - o Order ID, Customer ID, Product List, Total Amount, Status

- **Notifications**
  - o Notification ID, User ID, Message

ER Diagram for E-Shopp

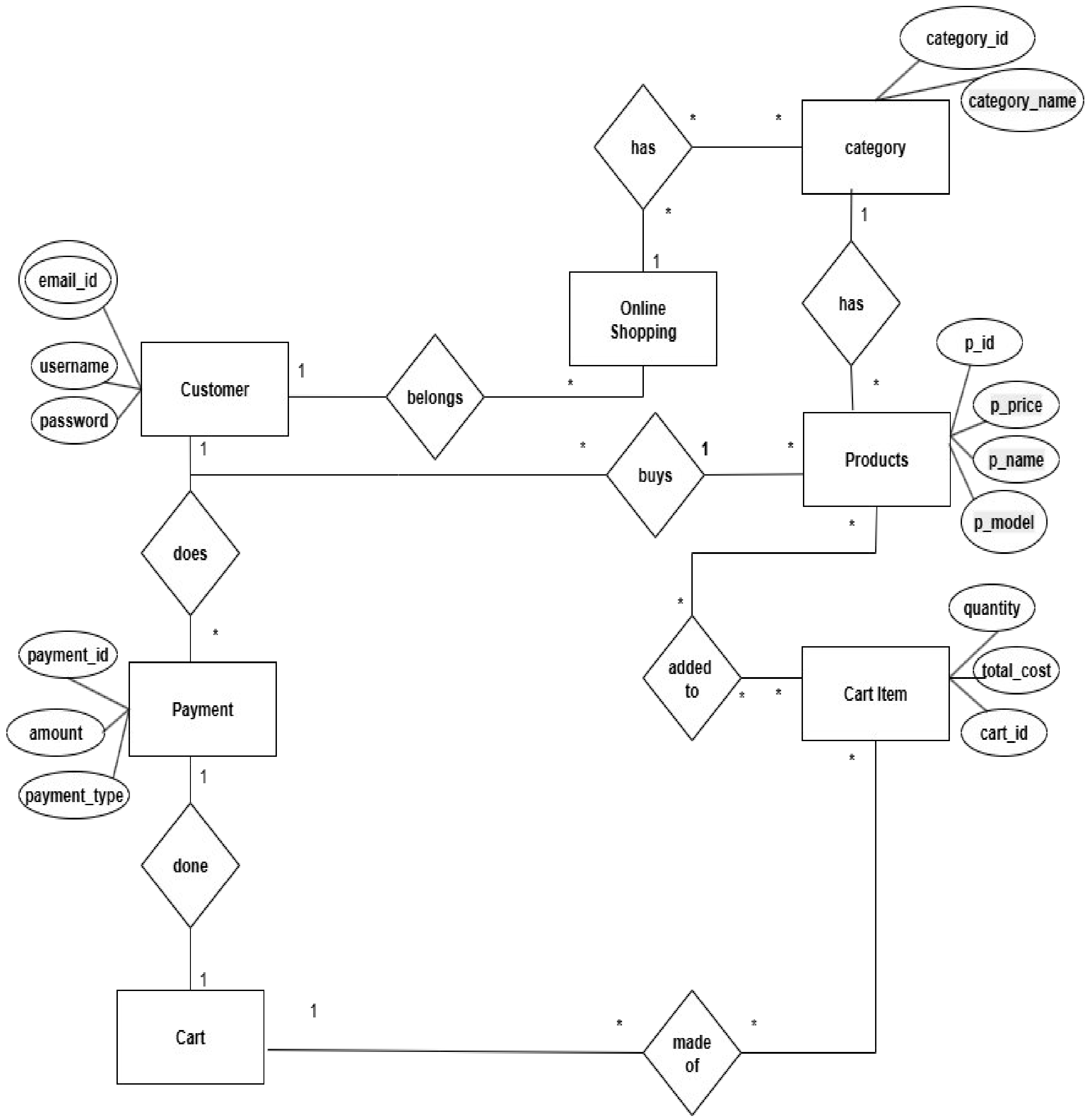

Figure 3: ER Diagram for E-Shopp

b. Process Modelling – DFD (Data Flow Diagrams)

Level-0 DFD: High-Level Overview

- **Main processes:**
  - o Manage Users
  - o Manage Products
  - o Place Order
  - o Process Payment (optional)

- o Manage Notifications

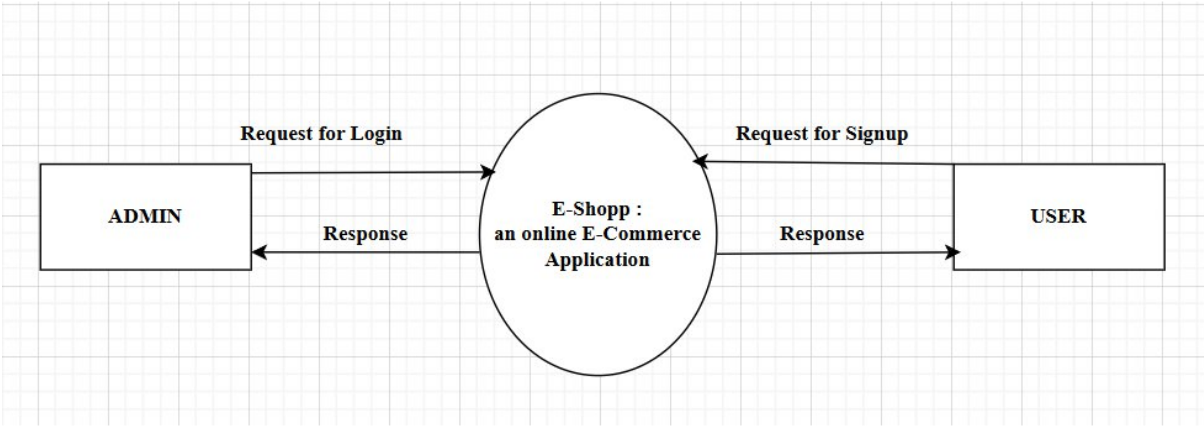

Figure 4: Level-0 DFD – E-Shopp

Level-1 DFD

Admin Side DFD - Level-1

- Request for Signup/Login
- Manage Products
- Manage Order
- View / Place Order
- View Report

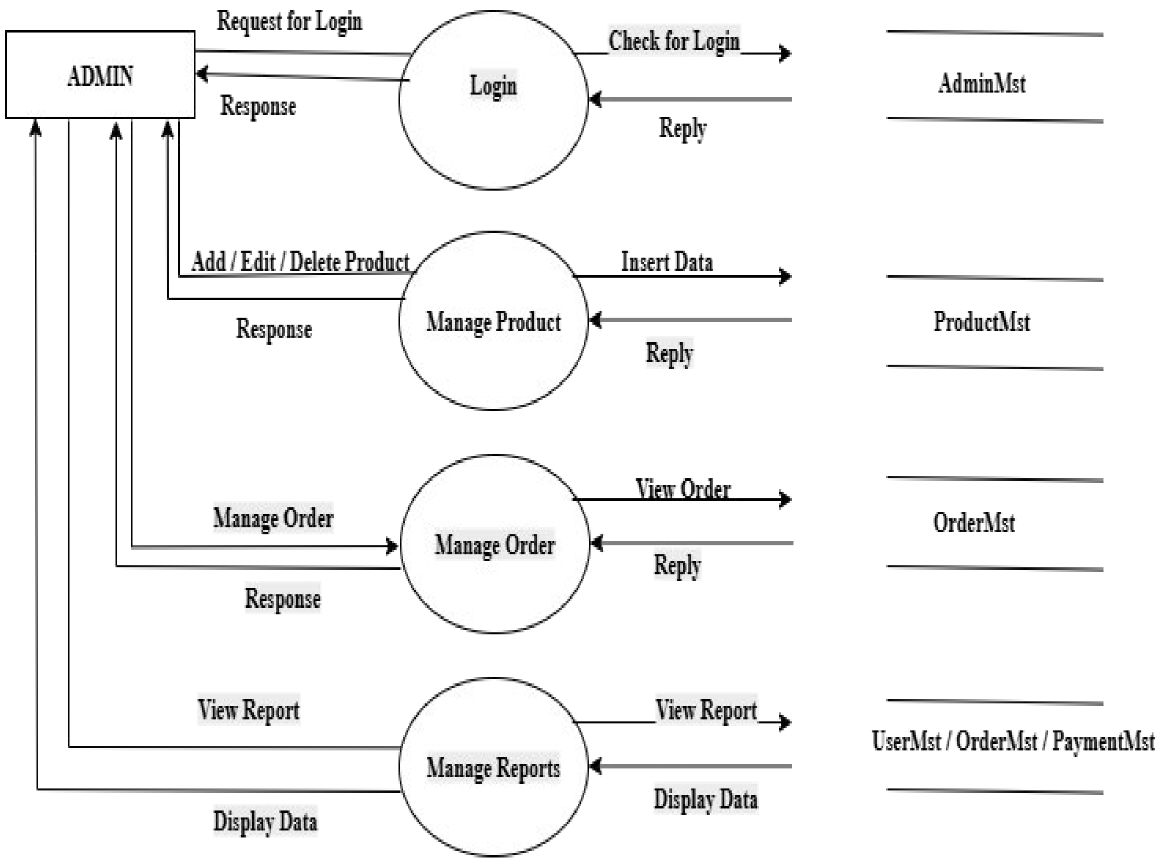

Figure 5 : Admin Side DFD - Level-1

Admin Side DFD - Level-2 (3.0)

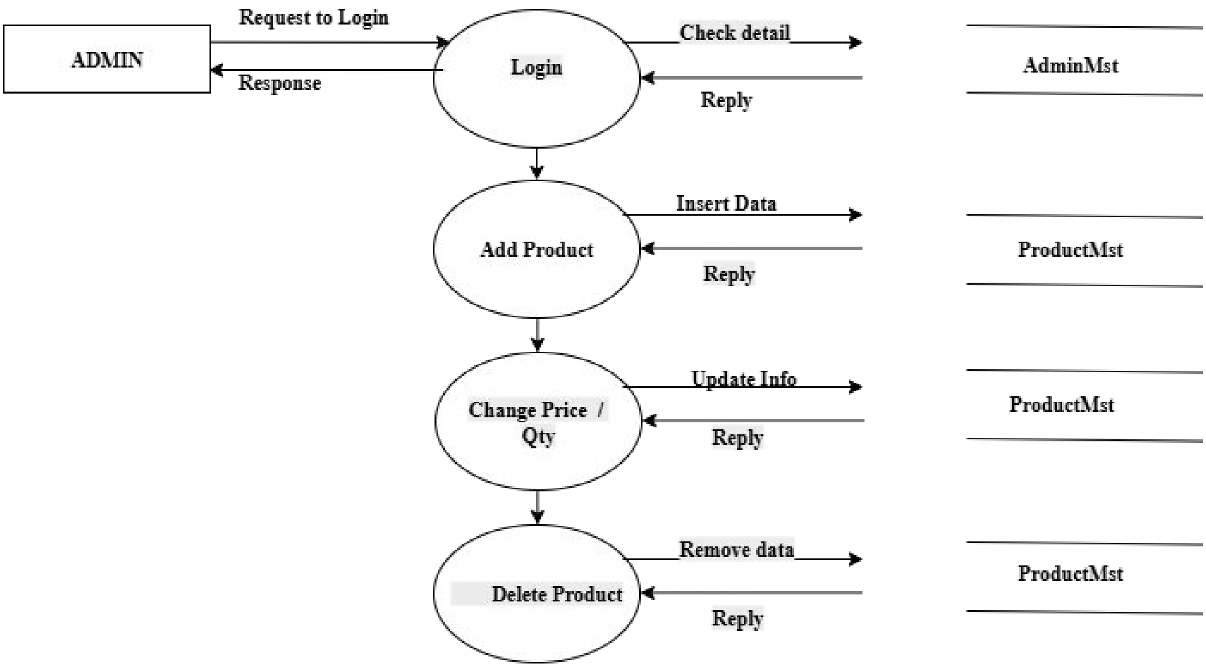

Figure 6 : Admin Side DFD - Level-2 (3.0)

Admin Side DFD - Level-2 (4.0)

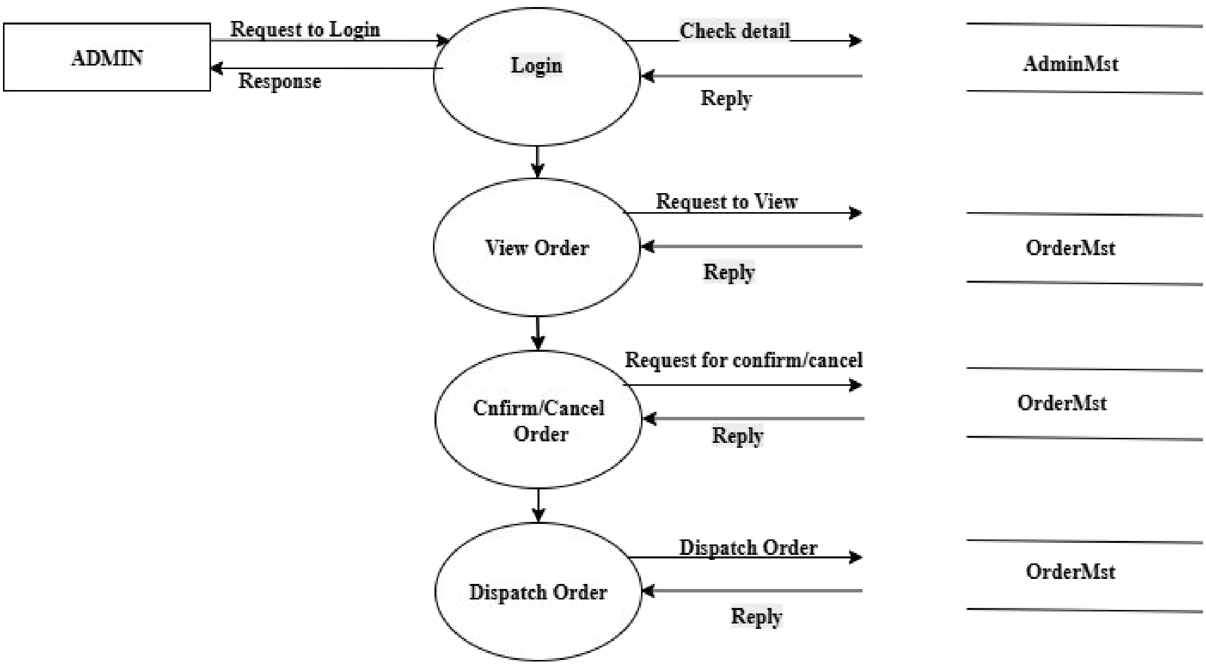

Figure 7 : Admin Side DFD - Level-2 (4.0)

Level-2 DFD

- Signup/Login
- Add/Edit/Delete Product
- Manage Orders

- View Dashboard

User Side DFD - Level-1

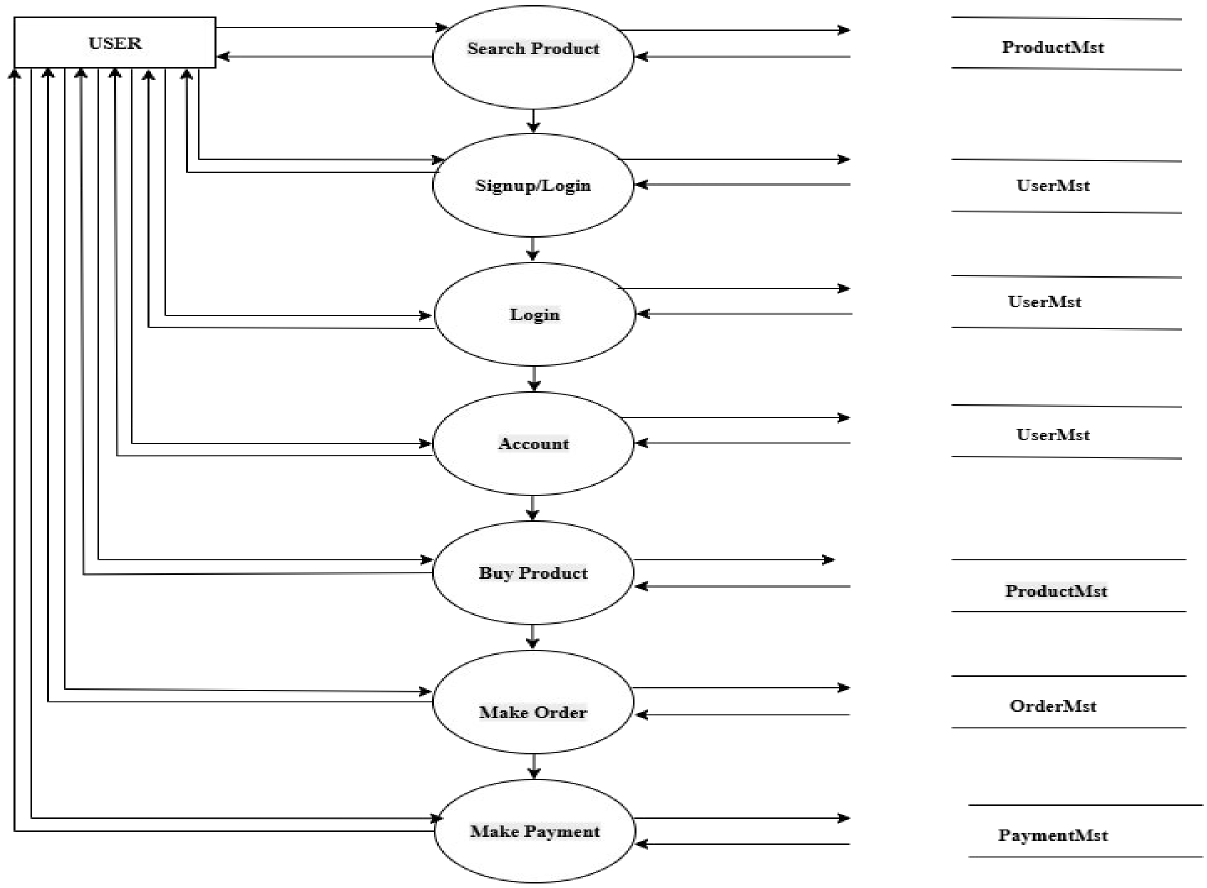

Figure 8 : User Side DFD - Level-1

User Side DFD - Level-2 (5.0)

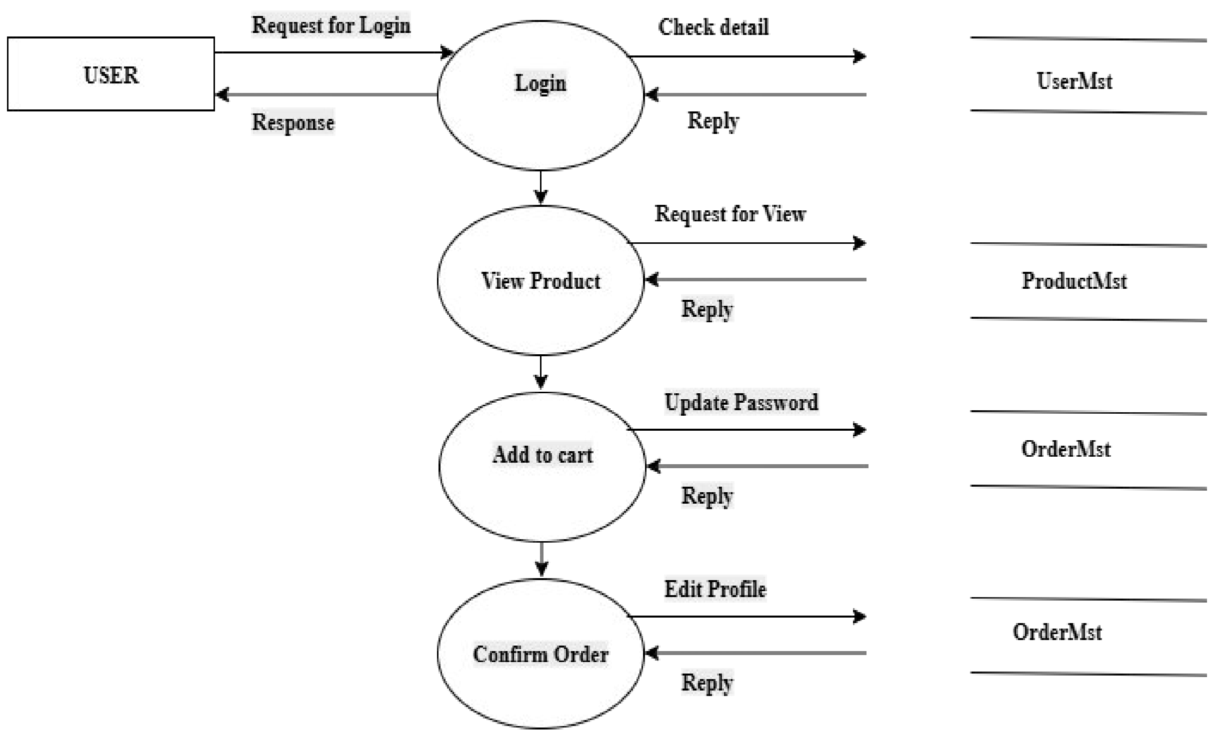

Figure 9 : User Side DFD - Level-2 (5.0)

Chapter 4: System Design

4.1 Design

System design is the structured process of defining the architecture, data structure, interface components, and flow of a system that full fills the functional and non-functional requirements. In E-Shopp, the design approach is user-centric, modular, and optimized for real-time performance using Firebase.

The frontend design focuses on clean user experience using Tailwind CSS, Bootstrap, and JavaScript, while the backend leverages Firebase Fire store (for database), Firebase Authentication, and Firebase Hosting.

4.1.1 Database Design

Since Firebase is a NoSQL cloud-based database, data is stored in the form of collections and documents, not traditional tables. Below is the high-level database schema design for E-Shopp:

🔑 **Entities (Collections)**

1. **Users**
   - Fields: user Id, name, email, role (customer/seller), address, profile Image, created At
   - Roles: Used to differentiate customers and sellers for role-based interfaces.
2. **Products**
   - Fields: productId, title, description, price, imageURL, category, stock, seller Id, created At
   - Relationships: Linked to a seller via seller Id.
3. **Orders**
   - Fields: order Id, customer Id, product Items (array of objects), total Amount, order Status, created At
   - Relationships: Connected to both users (customer) and products.
4. **Carts**
   - Fields: cart Id, user Id, items (array of {product Id, quantity}), updated At
   - Each customer has one active cart at a time.
5. **Notifications**
   - Fields: notification , user Id, type, message, is Read, timestamp
   - Sent to users for events like order placed, shipped, etc.
6. **Reviews**

    - Fields: review Id, product Id, user Id, rating, comment, created

🔁 Relationships

- A seller can list multiple products (1:N)
- A customer can place multiple orders (1:N)
- An order contains multiple products (1:N embedded array)
- A customer has one cart (1:1)
- A product can have many reviews (1:N)
- Notifications are user-specific (1:N between Users and Notifications)

✅ Design Guidelines Used

- Data Normalization: Firebase encourages embedding over referencing when data is read more often than written. E.g., order details store a snapshot of the product at the time of purchase.
- Security Rules: Role-based access control using Firebase Security Rules to restrict read/write access.
- Indexing: Fire store supports indexing by default; optimized queries using compound indexes for category + price, etc.

□ Example: Fire Store Structure

```
users/
  userId123/
    name: "Amit"
    role: "customer"
products/
  prod001/
    title: "Nike Shoes"
    price: 4500
    seller Id: "userId456"
orders/
  order987/
    customer Id: "userId123"
    product Items: [{product Id: "prod001", quantity: 2}]
    total Amount: 9000
carts/
  userId123/
```

items: [{product Id: "prod001", quantity: 1}]

notifications/

noti001/

user Id: "userId456"

message: "New order received!"

If Structured approach:

Database Design: Transformation of ER to relations and normalizations

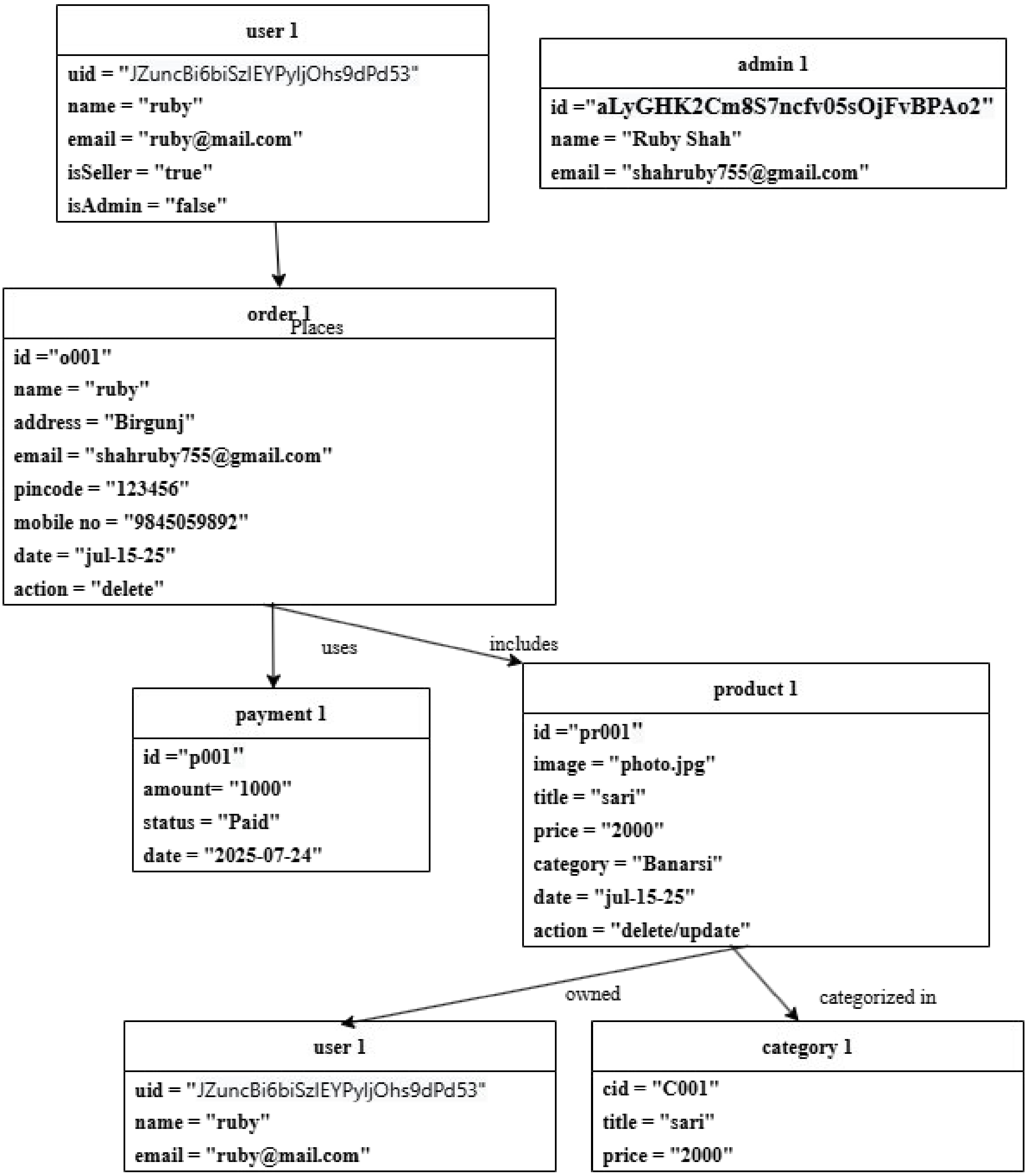

Figure 7 : Database Design: Transformation of ER to relations and normalizations

Forms and Report Design

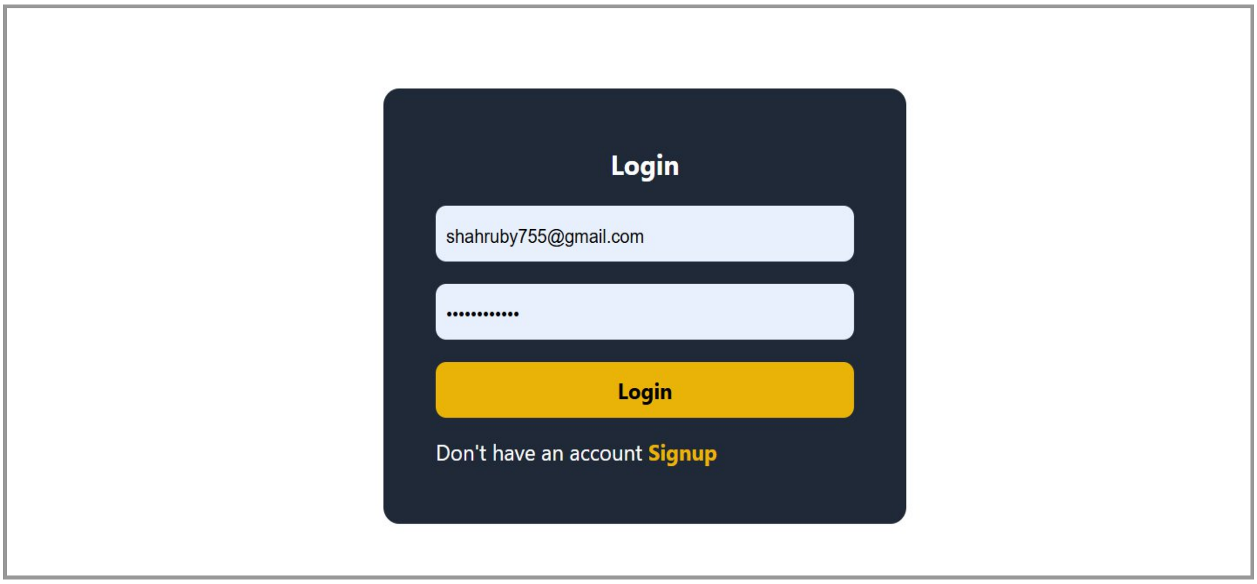

Figure 8 : Forms and Report Design

Interface and Dialogue Design

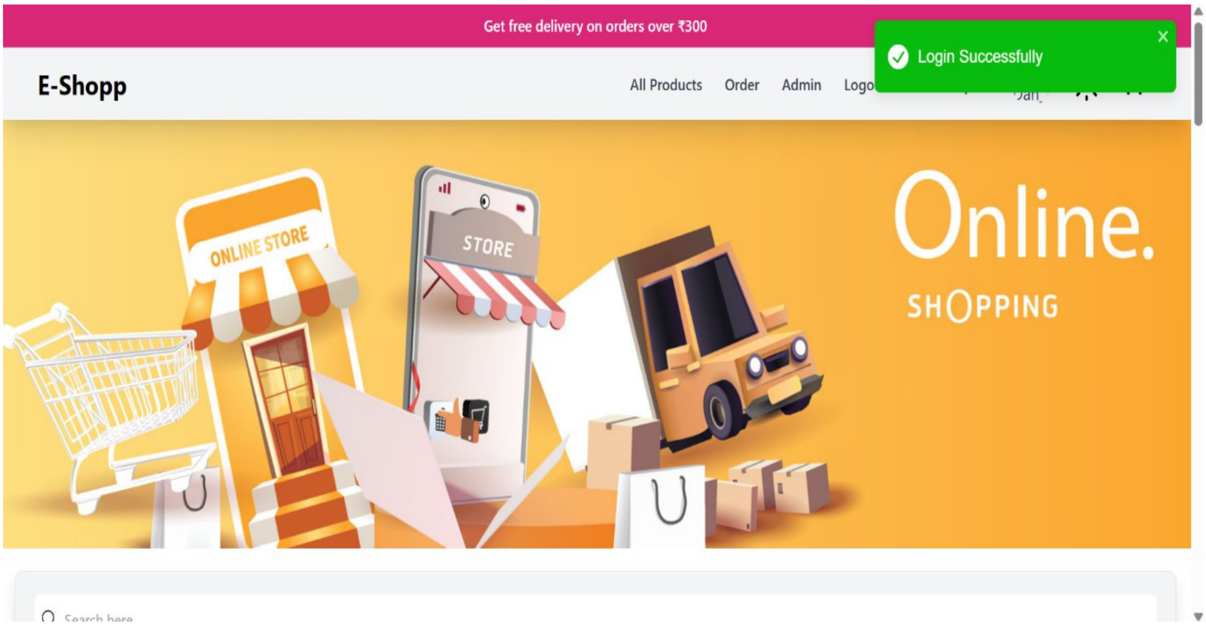

Figure 9 : Interface and Dialogue Design

4.2 Algorithm Details

COMMON FLOW FOR CRUD OPERATIONS

1. Users

- Add User (Signup):
  - o Input: User details (name, email, password, etc.)
  - o Process: Validate input → Check if email exists → Hash password → Save to DB

  - Output: Success/Failure message
- Update User:
  - Input: Authenticated user + new details
  - Process: Validate → Update fields in DB
  - Output: Success/Failure
- Delete User:
  - Input: Authenticated user ID
  - Process: Authenticate → Remove user record from DB
  - Output: Confirmation

2. Products

- Add Product (Admin):
  - Input: Product details (name, price, image, etc.)
  - Process: Validate → Save to DB
  - Output: Product ID
- Update Product:
  - Input: Product ID + New data
  - Process: Check if product exists → Update fields
  - Output: Success/Failure
- Delete Product:
  - Input: Product ID
  - Process: Verify product ownership → Delete
  - Output: Confirmation

3. View All Products

- Input: (Optional filters)
- Process: Fetch from DB, apply filters/pagination
- Output: List of products

4. Cart

- Add to Cart:
  - Input: User ID + Product ID + Quantity
  - Process: Check product availability → Update/Add to cart
  - Output: Confirmation
- Update Cart:
  - Input: Cart ID + new quantity
  - Process: Validate quantity → Update DB

- Output: Success

- Delete from Cart:
  - Input: Cart Item ID
  - Process: Remove item
  - Output: Confirmation

5. Orders

- Place Order:
  - Input: Cart + Payment info
  - Process: Validate cart → Process payment → Create order → Clear cart
  - Output: Order confirmation

6. Payments

- Process Payment:
  - Input: Card/UPI info, amount
  - Process: External API call (e.g., Stripe/PayPal) → Confirm payment
  - Output: Payment status

7. Notifications

- Trigger Notifications:
  - Input: Event (order placed, payment success, etc.)
  - Process: Create and queue notification (email, SMS, in-app)
  - Output: Notification sent

8. Login/Signup

- Signup:
  - Input: User details
  - Process: Validate + Save to DB
  - Output: Token + Success
- Login:
  - Input: Email & Password
  - Process: Verify credentials → Generate JWT/token
  - Output: Token + Success

Chapter 5: Implementation and Testing

5.1 Implementation

The **implementation phase** marks the transition of the system from design to reality. The development of **E-Shopp – An Online E-Commerce Application** followed a **prototyping model**, enabling frequent feedback and rapid iteration. Firebase was chosen for backend operations due to its real-time capabilities, while frontend development prioritized responsive design and user accessibility.

5.1.1 Tools Used

Category	**Tools/Technologies**
Frontend	HTML, Tailwind CSS, JavaScript, Bootstrap
Backend	Firebase (Fire store, Firebase Authentication, Firebase Hosting, Firebase Functions)
Development Environment	Visual Studio Code
Other Tools	GitHub (Version Control), draw.io (Diagram Design)

5.1.2 Implementation details of Modules

1. Authentication Module

- Uses **Firebase Authentication** to register/login users (customers and sellers).
- Google Sign-In and email-password login supported.
- Role-based access handled through user metadata.

2. Product Management

- Sellers can **add, update, or delete** products.
- Product fields: title, description, price, category, image URL, stock, seller Id.
- Images uploaded to **Firebase Storage**, links stored in Fire store.

3. Shopping Cart Module

- Logged-in users can **add/remove items** from their cart.
- Cart data stored in Fire store under the user’s document.
- Quantity updates are reflected in real-time using Firebase listeners.

4. Order Management Module

- Customers place orders from their cart.
- Orders are stored with status: pending, shipped, or delivered.
- Each order includes timestamp, customer Id, and product details (snapshot-based).

5. Notifications System

- Uses Firebase’s real-time database and **Cloud Functions** to push order updates.

- Customers get alerts when order status changes.

6. User Profile Management

- Customers/sellers can update their name, address, and avatar.
- Real-time sync with Fire store ensures consistency across sessions.

7. Search and Filter Functionality

- Search by keyword, category, price range.
- Fire store indexing improves performance on filter queries.

8. Admin Panel (Optional Enhancement)

- Admins can view registered users, manage reported items, and moderate reviews.

5.2 Testing

5.2.1 Test Cases for Unit Testing

Unit tests focus on individual functions, methods, or components, typically tested by developers.

1. User Authentication Module

- Test login with valid credentials.
- Test login with invalid password.
- Test login with empty email or password.
- Test password encryption method.
- Test session timeout logic.

2. Product Catalog Module

- Test fetching product list from database.
- Test filtering products by category.
- Test product detail function for a valid product ID.
- Test search functionality for case-insensitive match.
- Test pagination logic.

3. Shopping Cart Module

- Test adding an item to the cart.
- Test updating quantity of an item.
- Test removing an item.
- Test calculating total cart value.
- Test applying valid discount code.

4. Checkout and Payment Module

- Test order summary calculation logic.

- Test inventory deduction logic after order placed.
- Test payment processing function with mock payment API.
- Test validation of shipping address.
- Test order confirmation email generation function.

5. Order History Module

- Test retrieval of orders for a specific user.
- Test status update function (pending, shipped, delivered).
- Test order cancellation logic.

5.2.2 Test Cases for System Testing

System tests validate the entire application end-to-end. Typically conducted by QA testers.

1. User Registration and Login

- Verify user can register with valid input.
- Verify user cannot register with an already registered email.
- Verify user can log in with correct credentials.
- Verify user is shown error with wrong login details.

2. Browsing and Searching Products

- Verify user can view product categories.
- Verify filters (brand, price range, ratings) work as expected.
- Verify search returns relevant products.
- Verify user can view product details (image, description, price).

3. Cart Management

- Verify items can be added to cart from product page.
- Verify quantity can be updated in cart.
- Verify removing items updates total cost.
- Verify cart retains items when user logs out and back in.

4. Checkout Process

- Verify user can proceed to checkout with valid cart.
- Verify address selection and payment method selection work.
- Verify payment gateway redirects and processes correctly.
- Verify successful order shows confirmation page.

5. Order History and Tracking

- Verify user can view previous orders.
- Verify status updates reflect correctly (e.g., shipped, delivered).
- Verify tracking link opens in a new tab.

6. Security and Access Control

- Verify that restricted pages redirect unauthenticated users.
- Verify session expires after inactivity.
- Verify no SQL injection or XSS vulnerabilities on forms.

7. Mobile Responsiveness

- Verify layout works correctly on mobile/tablet devices.
- Verify buttons and links are clickable and visible on all screens.

8. Email Notifications

- Verify registration email is received.
- Verify order confirmation email is sent with correct details.
- Verify password reset email is triggered correctly.

Types of Testing Conducted

1. Unit Testing

Each module and component (cart, login, order) was tested in isolation.

2. Integration Testing

Tested interactions between authentication, product, and cart modules.

3. System Testing

Full end-to-end user flow tested from registration to order placement and confirmation.

4. User Acceptance Testing (UAT)

Involved test users from both buyer and seller roles. Feedback was used to improve navigation, form validations, and real-time feedback.

Sample Test Case Table

TC ID	Test Case	Steps	Expected Output	Result
TC-01	User Login	Enter valid email and password	Redirect to dashboard	✅ Pass
TC-02	Add to Cart	Select product and click "Add"	Product appears in cart	✅ Pass
TC-03	Place Order	Click "Checkout" with valid address	Order confirmation shown	✅ Pass
TC-04	Search Product	Enter keyword and search	Matching products shown	✅ Pass
TC-05	Signup User	Fill signup form and submit	Account created	✅ Pass
TC-06	Product Upload (Seller)	Fill form and submit	Product listed on homepage	✅ Pass
TC-07	Logout	Click logout button	Redirect to login page	✅ Pass

5.3 Result Analysis

Test Results Summary

Module	Tested Components	Status
Authentication	Signup, Login, Logout, Role	✅ Pass
Cart	Add, Remove, Update Quantity	✅ Pass
Orders	Create, Update Status	✅ Pass
Search	Keyword, Category Filter	✅ Pass
Firebase Storage	Product Image Upload	✅ Pass

Chapter 6: Conclusion and Future Recommendations

6.1 Conclusion

The development of **E-Shopp** successfully delivers a robust, scalable, and responsive e-commerce platform focused on local business and user-friendly digital retailing. The use of Firebase allowed for real-time functionality, secure user authentication, and seamless integration across services. All core modules were tested rigorously, ensuring reliability and usability for both customers and sellers.

6.2 Future Recommendations

To ensure continued success, user engagement, and scalability, here are suggested improvements:

🔧 1. Enhanced UX/UI

- **Mobile App Version** using Flutter or React Native for better mobile usability.
- **Multilingual Support**: Add Nepali and Hindi language options.

🤖 2. Smart Search & Personalization

- Implement **AI-based product recommendations**.
- Use **customer purchase behaviour** to personalize homepages.

🔄 3. Real-Time Features

- **Live Order Status Updates** using Firebase Functions.
- **Real-Time Stock Updates** when users are viewing limited-quantity items.

📢 4. Communication Enhancements

- **In-App Messaging** between buyers and sellers.
- **Automated Email Notifications** for order confirmations, shipping, etc.

🔐 5. Account Security

- **Two-Factor Authentication (2FA)** for login.
- **Biometric Login** (optional via PWA standards).

📍 6. Geo-Based Features

- **Radius-Based Product Listings** using Google Maps API.
- Enable **location tagging** for sellers to enable local pickup/delivery.

💸 7. Payment Gateway Integration

- Integrate **eSewa**, **Khalti**, or **Stripe** for seamless checkout experience.

📈 8. Analytics and Insights

- Admin dashboard with insights on **popular categories**, **top-selling items**, and **buyer activity**.

□ 9. Review & Rating System

- Let users **rate products and sellers**.
- Implement **spam detection** to prevent abuse.

☁ 10. Cloud Scalability

- Host on **Firebase Hosting** and optionally migrate to **Google Cloud Run** or **Ver cel** for advanced scaling.
- Set up **automatic backups** for Fire store.

📚 Citation & Referencing

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In-Text Citation Examples

- Firebase provides real-time data synchronization for seamless cart updates.
- Bootstrap simplifies layout and responsiveness.
- Tailwind's utility-first approach allows rapid frontend prototyping.